

Linux Assessment

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Question:

Duration : 2 Hours

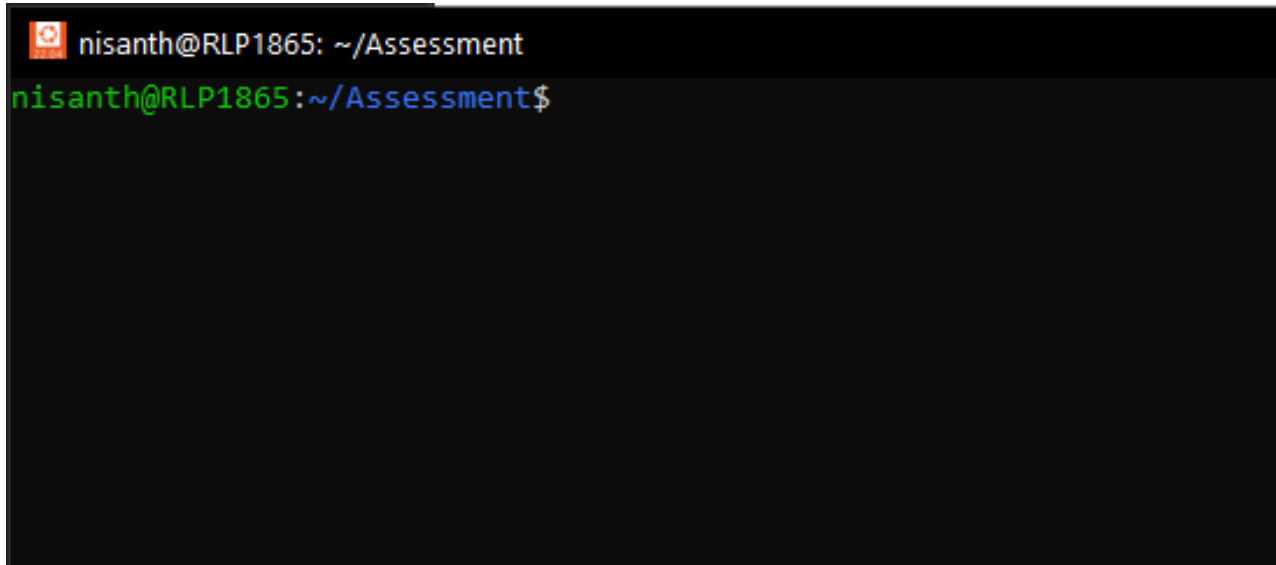
Task:

Prerequisites : Insall WSL Extension is your VsCode installed on Ubuntu

Note : Take Screen Shots of every step and document it,

1. Open the WSL terminal.
2. Create folder called LabExercise
3. Rename the LabExercise directory to LabWork.
4. Move files form old_directory_name to new_directory_name.
5. Update package lists using Linux Shell Commands.
6. Upgrade packages using Linux Shell Commands.
7. Navigate to your project folder using Linux Shell Commands. Open and type vim to create or edit a file.
8. Use Vim commands to edit and save the file. Use nano commands to edit and save the file.
9. Open the folder in Visual Studio Code run code. Develop C# console code for displaying list of Employees data using .net Core 8.0.
10. List all files, including hidden ones using Linux Shell Commands.

Open the WSL terminal

A screenshot of a Windows Subsystem for Linux (WSL) terminal window. The title bar at the top shows a red icon with a white 'x' and the text 'nisanth@RLP1865: ~/Assessment'. The terminal content shows the prompt 'nisanth@RLP1865:~/Assessment\$' in green text on a black background.

```
nisanth@RLP1865: ~/Assessment  
nisanth@RLP1865:~/Assessment$
```

Create folder called LabExercise

```
nisanth@RLP1865: ~/Assessment
nisanth@RLP1865:~/Assessment$ mkdir LabExercise
nisanth@RLP1865:~/Assessment$ ls
LabExercise
nisanth@RLP1865:~/Assessment$ _
```

The command `mkdir` allows the user to create an empty directory in the current working directory.

Rename the LabExercise directory to LabWork

```
nisanth@RLP1865: ~/Assessment
nisanth@RLP1865:~/Assessment$ mv LabExercise LabWork.
nisanth@RLP1865:~/Assessment$ ls
LabWork.
nisanth@RLP1865:~/Assessment$ mv Labwork. Labwork
mv: cannot stat 'Labwork.': No such file or directory
nisanth@RLP1865:~/Assessment$ mv LabWork. LabWork
nisanth@RLP1865:~/Assessment$ ls
LabWork
nisanth@RLP1865:~/Assessment$ _
```

As the directory “LabWork” doesn’t exist, the mv command simply renames the “LabExercise” directory.

But, if a directory specified in the “new_directory” of the command’s syntax, then the mv command would have moved the entire directory into the specified directory.

Move files form old_directory_name to new_directory_name

***Moving Directory:** The entire directly is being moved into a pre-existing directory.

```
nisanth@RLP1865: ~/Assessment/NewLab/LabWork
nisanth@RLP1865:~/Assessment$ ls
LabWork
nisanth@RLP1865:~/Assessment$ cd LabWork/
nisanth@RLP1865:~/Assessment/LabWork$ touch example.txt
nisanth@RLP1865:~/Assessment/LabWork$ ls
example.txt
nisanth@RLP1865:~/Assessment/LabWork$ cd ..
nisanth@RLP1865:~/Assessment$ mkdir NewLab
nisanth@RLP1865:~/Assessment$ mv LabWork NewLab
nisanth@RLP1865:~/Assessment$ ls
NewLab
nisanth@RLP1865:~/Assessment$ cd NewLab/
nisanth@RLP1865:~/Assessment/NewLab$ ls
LabWork
nisanth@RLP1865:~/Assessment/NewLab$ cd LabWork/
nisanth@RLP1865:~/Assessment/NewLab/LabWork$ ls
example.txt
nisanth@RLP1865:~/Assessment/NewLab/LabWork$ _
```

***Moving A File Alone:** The specific file alone is being moved into a pre-existing directory.

```
nisanth@RLP1865: ~/Assessment/LabWork/NewLab
nisanth@RLP1865:~/Assessment$ ls
LabWork
nisanth@RLP1865:~/Assessment$ cd LabWork/
nisanth@RLP1865:~/Assessment/LabWork$ touch example.txt
nisanth@RLP1865:~/Assessment/LabWork$ ls
example.txt
nisanth@RLP1865:~/Assessment/LabWork$ mkdir NewLab
nisanth@RLP1865:~/Assessment/LabWork$ ls
NewLab example.txt
nisanth@RLP1865:~/Assessment/LabWork$ mv example.txt NewLab
nisanth@RLP1865:~/Assessment/LabWork$ ls
NewLab
nisanth@RLP1865:~/Assessment/LabWork$ cd NewLab/
nisanth@RLP1865:~/Assessment/LabWork/NewLab$ ls
example.txt
nisanth@RLP1865:~/Assessment/LabWork/NewLab$
```

Update package lists using Linux Shell Commands

```
nisanth@RLP1865: ~/Assessment
nisanth@RLP1865:~/Assessment$ sudo apt update
[sudo] password for nisanth:
Hit:1 http://archive.ubuntu.com/ubuntu jammy InRelease
Get:2 http://archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:3 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Get:4 http://archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:5 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [2453 kB]
Get:6 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [2208 kB]
Get:7 http://archive.ubuntu.com/ubuntu jammy-updates/main Translation-en [403 kB]
Get:8 http://archive.ubuntu.com/ubuntu jammy-updates/restricted amd64 Packages [3241 kB]
Get:9 http://archive.ubuntu.com/ubuntu jammy-updates/restricted Translation-en [574 kB]
Get:10 http://security.ubuntu.com/ubuntu jammy-security/main Translation-en [340 kB]
Err:9 http://archive.ubuntu.com/ubuntu jammy-updates/restricted Translation-en
Hash Sum mismatch
Hashes of expected file:
- Filesize:574384 [weak]
- SHA256:7d4434118592d0b14c9b203e293c0f96f34c1f13a1d43f3aa621e00c17689e61
```

Upgrade packages using Linux Shell Commands

```
nisanth@RLP1865: ~/Assessment
nisanth@RLP1865:~/Assessment$ sudo apt upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
nisanth@RLP1865:~/Assessment$ _
```

Navigate to your project folder using Linux Shell Commands. Open and type vim to create or edit a file

Navigation to desired directory:

```
Select nisanth@RLP1865: ~/Assessment/LabWork
nisanth@RLP1865:~$ ls
Assessment  Training
nisanth@RLP1865:~$ cd Assessment/LabWork/
nisanth@RLP1865:~/Assessment/LabWork$
```

Using VIM editor:

If the specified filename doesn't exist in the directory, a new file in the said name will be created.

```
nisanth@RLP1865: ~/Assessment/LabWork
nisanth@RLP1865:~$ ls
Assessment  Training
nisanth@RLP1865:~$ cd Assessment/LabWork/
nisanth@RLP1865:~/Assessment/LabWork$ vim sample.txt_
```

```
nisanth@RLP1865: ~/Assessment/LabWork
This is a sample text, entered from the vim text editor._
~
~
~
~
~
~
~
```

Press “Esc” to exit INSERT mode. Type in “:wq” to write out and quit the editor,
The specified command will be shown in the bottom left corner of your shell.
This helps in identifying which mode the VIM editor is in.


```
nisanth@RLP1865: ~/Assessment/LabWork
nisanth@RLP1865:~$ ls
Assessment  Training
nisanth@RLP1865:~$ cd Assessment/LabWork/
nisanth@RLP1865:~/Assessment/LabWork$ vim sample.txt
nisanth@RLP1865:~/Assessment/LabWork$ cat sample.txt
This is a sample text, entered from the vim text editor.
nisanth@RLP1865:~/Assessment/LabWork$
```

Editing using VIM editor:

But if the file does exist, then it counts as editing a pre-existing file, and doesn't create a new one without any purpose.

[illegible]

```
nisanth@RLP1865: ~/Assessment/LabWork
nisanth@RLP1865:~$ ls
Assessment  Training
nisanth@RLP1865:~$ cd Assessment/LabWork/
nisanth@RLP1865:~/Assessment/LabWork$ vim sample.txt
nisanth@RLP1865:~/Assessment/LabWork$ cat sample.txt
This is a sample text, entered from the vim text editor.
nisanth@RLP1865:~/Assessment/LabWork$ vim sample.txt
nisanth@RLP1865:~/Assessment/LabWork$ cat sample.txt
This is a sample text, entered from the vim text editor
This a simple edit..
nisanth@RLP1865:~/Assessment/LabWork$
```

Use Vim commands to edit and save the file. Use nano commands to edit and save the file

Using VIM:

If the specified filename doesn't exist in the directory, a new file in the said name will be created.

```
nisanth@RLP1865: ~/Assessment/LabWork
nisanth@RLP1865:~$ ls
Assessment  Training
nisanth@RLP1865:~$ cd Assessment/LabWork/
nisanth@RLP1865:~/Assessment/LabWork$ vim sample.txt_
```

Press “i” to enter INSERT mode:

```
~
~
~
~
-- INSERT --
```

Start typing your text as the cursor appears.

```
nisanth@RLP1865: ~/Assessment/LabWork
This is a sample text, entered from the vim text editor._
~
~
~
~
```

Press “Esc” to exit INSERT mode. Type in “:wq” to write out and quit the editor:

```
~  
~  
~  
~  
~  
:wq
```

The specified command will be shown in the bottom left corner of your shell. This helps in identifying which mode the VIM editor is in.

```
nisanth@RLP1865: ~/Assessment/LabWork  
nisanth@RLP1865:~$ ls  
Assessment  Training  
nisanth@RLP1865:~$ cd Assessment/LabWork/  
nisanth@RLP1865:~/Assessment/LabWork$ vim sample.txt  
nisanth@RLP1865:~/Assessment/LabWork$ cat sample.txt  
This is a sample text, entered from the vim text editor.  
nisanth@RLP1865:~/Assessment/LabWork$
```

Using NANO:

If the specified filename doesn't exist in the directory, a new file in the said name will be created.

```
nisanth@RLP1865: ~/Assessment/LabWork
nisanth@RLP1865:~/Assessment/LabWork$ nano again.txt
nisanth@RLP1865:~/Assessment/LabWork$ _
```

Simply start writing:

Unlike the VIM editor, nano allows user to write as soon as the editor is opened. It contains a helping guide at the bottom to show the users the list of commands to edit and save their file. A very user friendly editor.

```
nisanth@RLP1865: ~/Assessment/LabWork
GNU nano 6.2                                again.txt *
This is the nano text editor._
```

Press “Ctrl + O” to write out:

```
File Name to Write: again.txt
^G Help                                M-D DOS Format          M-A Append
^C Cancel                              M-M Mac Format          M-P Prepend
```

Press “Ctrl + X” to exit:

```
^G Help                                ^O Write Out            ^W Where Is             ^K Cut
^X Exit                                ^R Read File            ^\ Replace              ^U Paste
```



nisanth@RLP1865: ~/Assessment/LabWork

```
nisanth@RLP1865:~/Assessment/LabWork$ nano again.txt
nisanth@RLP1865:~/Assessment/LabWork$ cat again.txt
This is the nano text editor.
nisanth@RLP1865:~/Assessment/LabWork$ _
```

Editing:

But if the file does exist, then it counts as editing a pre-existing file, and doesn't create a new one without any purpose.



nisanth@RLP1865: ~/Assessment/LabWork

```
nisanth@RLP1865:~/Assessment/LabWork$ nano again.txt
nisanth@RLP1865:~/Assessment/LabWork$ cat again.txt
This is the nano text editor.
nisanth@RLP1865:~/Assessment/LabWork$ nano again.txt _
```



nisanth@RLP1865: ~/Assessment/LabWork

GNU nano 6.2

```
This is a simple edit._
This is the nano text editor.
```

Write out and Exit:



nisanth@RLP1865: ~/Assessment/LabWork

```
nisanth@RLP1865:~/Assessment/LabWork$ nano again.txt
nisanth@RLP1865:~/Assessment/LabWork$ cat again.txt
This is the nano text editor.
nisanth@RLP1865:~/Assessment/LabWork$ nano again.txt
nisanth@RLP1865:~/Assessment/LabWork$ cat again.txt
This is a simple edit.
This is the nano text editor.
nisanth@RLP1865:~/Assessment/LabWork$
```

Open the folder in Visual Studio Code run code. Develop C# console code for displaying list of Employees data using .net Core 8.0. :

Creating console app:

Upon using the said command a basic skeletal console app will be created in our current working directory.

```
nisanth@RLP1865: ~/Assessment/LabWork
nisanth@RLP1865:~/Assessment/LabWork$ dotnet new console
The template "Console App" was created successfully.

Processing post-creation actions...
Restoring /home/nisanth/Assessment/LabWork/LabWork.csproj:
  Determining projects to restore...
  Restored /home/nisanth/Assessment/LabWork/LabWork.csproj (in 1.6 sec).
Restore succeeded.

nisanth@RLP1865:~/Assessment/LabWork$ ls
LabWork.csproj  Program.cs  obj
nisanth@RLP1865:~/Assessment/LabWork$
```

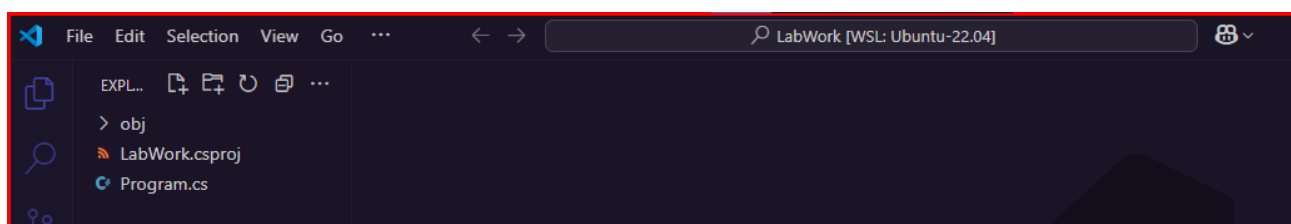
Opening VS Code:

```
nisanth@RLP1865: ~/Assessment/LabWork
nisanth@RLP1865:~/Assessment/LabWork$ dotnet new console
The template "Console App" was created successfully.

Processing post-creation actions...
Restoring /home/nisanth/Assessment/LabWork/LabWork.csproj:
  Determining projects to restore...
  Restored /home/nisanth/Assessment/LabWork/LabWork.csproj (in 1.6 sec).
Restore succeeded.

nisanth@RLP1865:~/Assessment/LabWork$ ls
LabWork.csproj  Program.cs  obj
nisanth@RLP1865:~/Assessment/LabWork$ code .
nisanth@RLP1865:~/Assessment/LabWork$
```

VS Code opened in the WSL environment:



Build the written code:

After being done with the code, the given command will create the compiled build_output folder for the console app to successfully run in Linux Environment.

```
nisanth@RLP1865: ~/Assessment/LabWork
nisanth@RLP1865:~/Assessment/LabWork$ dotnet build --output ./build_output
MSBuild version 17.8.19+c3ade832a for .NET
  Determining projects to restore...
  All projects are up-to-date for restore.
/usr/lib/dotnet/sdk/8.0.114/Current/SolutionFile/ImportAfter/Microsoft.NET.Sdk.Solution.targets(36,5): The "--output" option isn't supported when building a solution. Specifying a solution-level output projects copying outputs to the same directory, which can lead to inconsistent builds. [/home/nisanth/LabWork.sln]
/home/nisanth/Assessment/LabWork/Program.cs(15,32): warning CS8604: Possible null reference argument in 'int.Parse(string s)'. [/home/nisanth/Assessment/LabWork/LabWork.csproj]
/home/nisanth/Assessment/LabWork/Program.cs(29,40): warning CS8604: Possible null reference argument in 'int.Parse(string s)'. [/home/nisanth/Assessment/LabWork/LabWork.csproj]
/home/nisanth/Assessment/LabWork/Program.cs(31,35): warning CS8600: Converting null literal or possible null value to non-nullable type. [/home/nisanth/Assessment/LabWork/LabWork.csproj]
/home/nisanth/Assessment/LabWork/Program.cs(33,35): warning CS8600: Converting null literal or possible null value to non-nullable type. [/home/nisanth/Assessment/LabWork/LabWork.csproj]
/home/nisanth/Assessment/LabWork/Program.cs(34,51): warning CS8604: Possible null reference argument in 'Employee.Employee(int id, string name, string dept)'. [/home/nisanth/Assessment/LabWork/LabWork.csproj]
/home/nisanth/Assessment/LabWork/Program.cs(34,57): warning CS8604: Possible null reference argument in 'Employee.Employee(int id, string name, string dept)'. [/home/nisanth/Assessment/LabWork/LabWork.csproj]
LabWork -> /home/nisanth/Assessment/LabWork/build_output/LabWork.dll

Build succeeded.
```

Run the code:

Adding Employee details:

```
nisanth@RLP1865: ~/Assessment/LabWork
nisanth@RLP1865:~/Assessment/LabWork$ dotnet run

EMPLOYEE DETAILS
=====
1. ADD EMPLOYEE
2. DISPLAY EMPLOYEE
0. EXIT
=====

ENTER YOUR CHOICE:1

=====
ENTER ID:1
ENTER NAME:Nisanth
ENTER DEPT:Dev
EMPLOYEE ADDED SUCCESSFULLY
=====
```

Displaying Employee details:

```
EMPLOYEE DETAILS
=====
1. ADD EMPLOYEE
2. DISPLAY EMPLOYEE
0. EXIT
=====

ENTER YOUR CHOICE:2

=====
ID: 1 Name: Nisanth Dept: dev
ID: 2 Name: Bonsai Plays Dept: Game Dev
=====
```

Exiting Application:

```
EMPLOYEE DETAILS
=====
1. ADD EMPLOYEE
2. DISPLAY EMPLOYEE
0. EXIT
=====

ENTER YOUR CHOICE:0

=====
THANK YOU
=====
nisanth@RLP1865:~/Assessment/LabWork$
```


Actual Code

Employee.cs:

```
public class Employee
{
    public int EmpID { get; set; }
    public string Name { get; set; }
    public string Dept { get; set; }

    public Employee(int id, string name, string dept)
    {
        this.EmpID = id;
        this.Name = name;
        this.Dept = dept;
    }

    public static void DisplayEmp(Employee e)
    {
        Console.WriteLine($"ID: {e.EmpID} Name: {e.Name} Dept: {e.Dept}\n");
    }
}
```

Program.cs:

```
class Company
{
    public static void Main()
    {
        List<Employee> employees = new List<Employee>();

        checkpoint1:
        Console.WriteLine("\nEMPLOYEE DETAILS");
        Console.WriteLine("=====");
        Console.WriteLine("1. ADD EMPLOYEE");
        Console.WriteLine("2. DISPLAY EMPLOYEE");
        Console.WriteLine("0. EXIT");
        Console.WriteLine("=====");
        Console.Write("\nENTER YOUR CHOICE:");
        int choice = int.Parse(Console.ReadLine());
        switch (choice)
        {
            case 0:
            {
                Console.WriteLine("\n=====");
                Console.WriteLine("THANK YOU");
                Console.WriteLine("=====");
                return;
            }
            case 1:
            {
                Console.WriteLine("\n=====");
                Console.Write("ENTER ID:");
                int id = int.Parse(Console.ReadLine());
                Console.Write("ENTER NAME:");
                string name = Console.ReadLine();
                Console.Write("ENTER DEPT:");
                string dept = Console.ReadLine();
                Employee e = new Employee(id, name, dept);
                employees.Add(e);
                Console.WriteLine("EMPLOYEE ADDED SUCCESSFULLY");
                Console.WriteLine("=====");
                goto checkpoint1;
            }
        }
    }
}
```

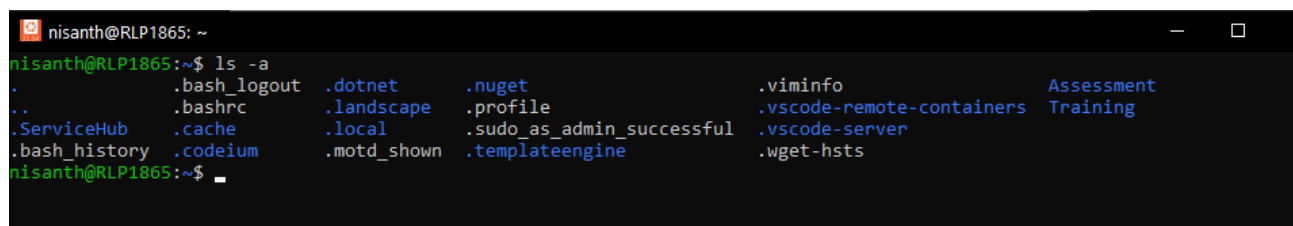
```

case 2:
{
    Console.WriteLine("\n=====");
    foreach (Employee e in employees)
    {
        Employee.DisplayEmp(e);
    }
    Console.WriteLine("=====");
    goto checkpoint1;
}
}
}

```

List all files, including hidden ones using Linux Shell Commands

Using the ls command with -a option:

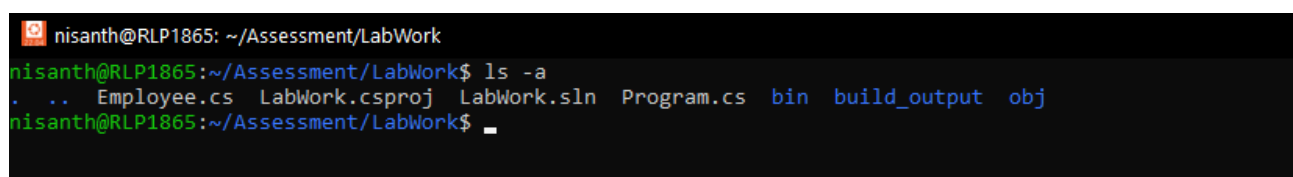


```

nisanth@RLP1865: ~
nisanth@RLP1865:~$ ls -a
.      .bash_logout  .dotnet      .nuget        .viminfo      Assessment
..     .bashrc       .landscape   .profile      .vscode-remote-containers Training
.ServiceHub .cache        .local       .sudo_as_admin_successful .vscode-server
.bash_history .codeium      .motd_shown  .templateengine .wget-hsts
nisanth@RLP1865:~$

```

*Note: No hidden files present in the project directory.



```

nisanth@RLP1865: ~/Assessment/LabWork
nisanth@RLP1865:~/Assessment/LabWork$ ls -a
.  .. Employee.cs LabWork.csproj LabWork.sln Program.cs bin build_output obj
nisanth@RLP1865:~/Assessment/LabWork$

```